

EXCLUSIVE

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**SCIENCE
&
TECHNOLOGY-13**



Landing on the Moon was last performed in 1972 by the Apollo 17 mission. Apollo 8 took humans around the Moon for the first time ever.

WHAT DO YOU KNOW ABOUT THE RECENT ARTEMIS II MISSION?

- Four astronauts blasted off aboard a massive NASA rocket on April 1, 2026, on a long-anticipated journey around the Moon, the first crewed lunar flyby in more than 50 years.
- The Artemis II mission uses the Space Launch System (SLS) rocket, and the crew capsule called Orion. The crew selected the name of "Integrity" to the spacecraft to signify the foundation of trust, candor, and humility, as well as the technical integration required for the mission.
- On April 6, 2026, Artemis II mission set the record for the longest human flyby reaching a maximum distance of 252,756 miles (406,771 km) from Earth. This achievement surpasses the 1970 Apollo 13 record of 248,655 miles.
- Earth's gravity pulls the spacecraft back to splash down in the Pacific Ocean.



- These four astronauts - three of USA and one from Canada - became the first humans to travel beyond low earth orbit since Apollo 17 in 1972.
- One of them became the first person of color, one the first woman and one, the first non-US citizen to embark on a lunar trajectory.
- Apollo, after whom the Moon missions between 1969 and 1972 were named, was the son of Zeus, the mythical Greek king of gods and Leto. More than half a century after the last Apollo mission, it is the turn of twin sister Artemis to lend her name to a Moon mission.
- The mission will check the life support and environment systems, communications and navigation systems as well as physiological and biological responses of the human body to deep space travel.

The capsule has to brave many difficulties such as fan belt radiation, unexpected solar storms and galactic cosmic rays and guard against even the minute errors in navigation that could lead to astray. Moreover, the speeded reentry will be far more than in a low earth orbit mission calling for ultra-advanced heat shields.



- In the older plan, the Artemis III mission was to land humans on the moon for the first time in the 21st century.
- In the new plan, Artemis III will launch a crewed Orion capsule to Earth's orbit to dock with prototype lunar landers designed by SpaceX and Blue Origin to ensure the technology works. Artemis III is planned for 2027.
- NASA will actually land American astronauts on the moon in the Artemis IV mission, currently planned for 2028.
- NASA also cancelled the Lunar Gateway project to build a space station orbiting the moon and reallocated its components to infrastructure that will eventually be installed on the moon's south pole.

ON THE OTHER HAND, AMBITIOUS PLANS OF CHINA



- China is set to conduct a test flight of its new Mengzhou crewed spacecraft this year. Its new and powerful Long March-10 rocket made its first low-altitude flight on February 11, 2026.
- The Chang'e 7 moon mission is also expected this year. It will explore the moon's south pole region for resources like water.
- The Chang'e 8 mission is expected around 2029 - it will have technologies to use resources on the moon itself, like a 3D-printer that will try to build structures using lunar soil.

China also plans to land humans on the Moon by 2030. It also has plans for its 'International Lunar Research Station' in the 2030s. It is designed to include research outposts, refueling depots, communication relays, and resource extraction sites.

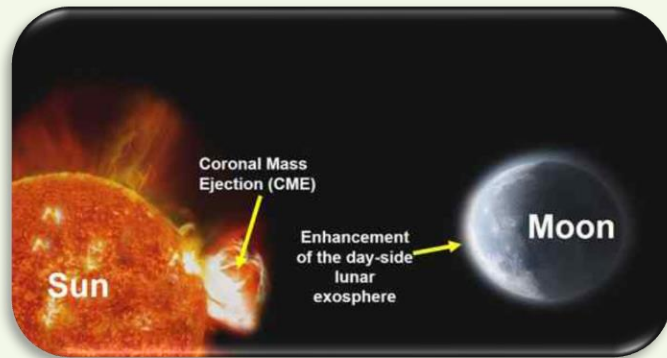
THERE ARE SEVERAL ADVANTAGES WITH THE OUTPOSTS OUTSIDE THE EARTH'S ATMOSPHERE!



- Studying radiation and collecting astronomical data are much easier.
- It could also be theoretically easier and cheaper to launch rockets from the Moon. Assembling spacecraft and other equipment may also be more efficient.
- Another major payoff may involve finding and exploiting resources, which may be abundant in space, while being scarce on the Earth, such as helium isotopes and rare earths. Simply finding significant quantities of water would be useful.
- Similar commercial equations apply to Mars and to comets and asteroids.

There could be huge extraterrestrial resources available to the first finders!

What do you know about the recent observations by lunar orbiter of Chandrayaan-2?



- The lunar orbiter of India's second Moon mission, Chandrayaan-2, has made the first-ever observation of the effects of the Sun's Coronal Mass Ejection (CME) on the Moon. The Chandrayaan-2's Atmospheric Composition Explorer-2 (CHACE-2) payload onboard the orbiter had made the observation.
- The observations from CHACE-2 showed an increase in the total pressure of the day side lunar atmosphere (very thin atmosphere), when the CME impacted the Moon.
- The increased quantity of solar coronal mass that impacted on the Moon enhanced the process of knocking off the atoms from the lunar surface, thereby liberating them to the lunar exosphere, which manifested as the enhancement of the total pressure in the sunlit lunar exosphere.

Launched on July 22, 2019, using the GSLV-MkIII-M1 rocket, Chandrayaan-2 carried eight experimental payloads.

VARIOUS REFORMS IN THE SPACE SECTOR DURING THE PAST FIVE YEARS SINCE 2020



- The State led reforms opened the Space Sector to private companies in 2023.
- The geospatial data was liberalized in 2021. The National Geospatial Policy 2022 aims to employ geospatial technology and data towards achieving Sustainable Development Goals, it aims to support innovation and creation in the field bridging the geospatial data divide. Guidelines were issued for acquiring and producing geospatial data and related services including maps.
- Indian National Space Promotion and Authorization Center (IN-SPACe) was created so as to ensure level playing field.
- Telecommunications Act, 2023 provided for satellite broadband services.
- On 21st February 2024, the Government opened the door to 100% FDI in the Space Sector.



1. In June 2025, Group Captain Shubhanshu Shukla displayed the tricolor above the International Space Station (ISS).
2. Chandrayaan-3 on August 23, 2023, made India the first nation to land near the lunar south pole. We celebrate National Space Day on August 23.
3. Chandrayaan-1 in 2008, confirmed the presence of water molecules, whereas Chandrayaan-2 in 2019, mapped the moon with high precision.
4. The Aditya-L1 mission of 2023, built through multi-institutional collaboration, is providing unprecedented insights into the Sun's corona and its impact on space weather.
5. XPoSat in 2024 is studying black holes, while SpaDeX in 2024 has demonstrated in-orbit docking for future space stations and lunar missions.
6. The upcoming programs include – Gaganyaan program for Human Space Flight, Chandrayaan-4 and 5 for deeper lunar exploration, a dedicated Venus mission, a Bharatiya Antariksha Station (BAS) by 2035, and an Indian human landing on the Moon by 2040.

IMPORTANT DEVELOPMENTS IN INDIA'S SPACE JOURNEY IN RECENT YEARS



Indian startups -Digantara, Pixxel, and XD-Links have launched satellites aboard SpaceX rockets.

7. The International Olympiad on Astronomy and Astrophysics was hosted in India in August 2025, and there were 300 participants from over 60 countries.
8. The South Asia satellite has provided neighbors with communication capacity, while during India's G-20 presidency in 2023, India announced a G-20 satellite for climate and environmental monitoring.
9. There are collaborative missions such as NASA-ISRO Synthetic Aperture Radar (NISAR) with NASA and Thermal Infrared Imaging Satellite for High-resolution Natural resource Assessment (TRISHNA) with CNES (French National Space Agency).
10. Other missions include Lunar Polar Exploration (LUPEX) with Japan Aerospace Exploration Agency (JAXA) and India's participation in the European Space Agency's PROBA-3.

1

CHANDRAYAAN-4

The Mission is expected to make India self-sufficient in technologies for manned missions and lunar sample analysis, with significant involvement from Indian industries and academic institutions.



- Chandrayaan-4 aims to develop and demonstrate technologies for returning to Earth after a successful landing on the Moon. It will also collect lunar samples for analysis back on Earth.
- ISRO will handle the development and launch of Chandrayaan-4, which is expected to be completed within 36 months.
- This mission will cost Rs. 2,104 crore. This budget includes spacecraft development, two LVM3 launches, deep space network support, and special tests.

2

VENUS ORBITER MISSION (VOM)

Venus is the Earth's hottest planet and is also called Earth's twin.

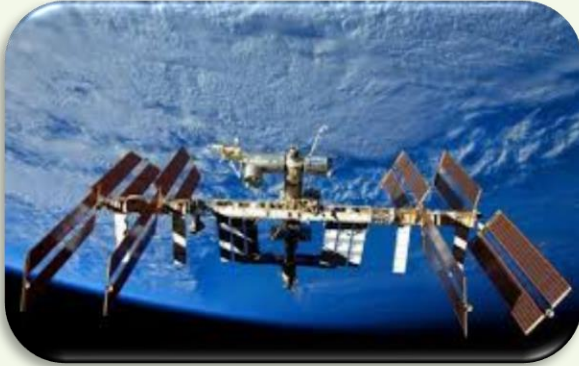


- The Venus Orbiter Mission (VOM) will focus on scientific exploration to better understand Venus's atmosphere and geology, generating extensive scientific data by probing its thick atmosphere.
- The Mission aims to orbit Venus with a scientific spacecraft to study the planet's surface, subsurface, atmospheric processes, and the Sun's impact on its atmosphere.
- Studying Venus is important because it is believed to have once been habitable like Earth.

The Mission is scheduled for launch in March 2028. The total budget for the Venus Orbiter Mission is Rs. 1,236 crore.

3

BHARATIYA ANTARIKSH STATION (BAS)



- At present, only two functioning space stations are there - US-led International Space Station and China's Tiangong. India will have its own space station for scientific research.
- The Cabinet approved the development of the first module of the Bharatiya Antariksh Station (BAS-1) and missions to validate technologies for building and operating BAS.
- The Gaganyaan programme will be revised to include these new developments and additional requirements, with a focus on completing eight missions by December 2028.
- The Gaganyaan programme, initially approved in December 2018, aims for human spaceflight to Low Earth Orbit and to support future Indian space explorations.
- India plans to operationalize Bharatiya Antariksh Station by 2035 and achieve a crewed lunar mission by 2040.

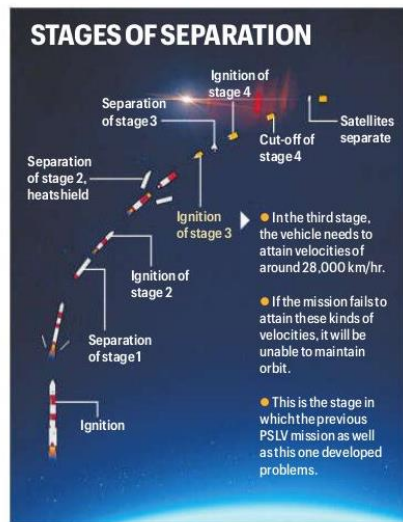
The revised Gaganyaan programme, under ISRO, has total enhanced funding of Rs. 20,193 crore.

4

NEXT GENERATION LAUNCH VEHICLE (NGLV)



- The Next Generation Launch Vehicle (NGLV) will offer three times the current payload capacity of the LVM3, at 1.5 times the cost. It is designed to carry up to 30 tonnes to Low Earth Orbit (LEO).
- India's existing launch vehicles, including PSLV, GSLV, LVM3, and SSLV, can launch satellites up to 10 tonnes to LEO and 4 tonnes to Geo-Synchronous Transfer Orbit (GTO). The NGLV will build on this capability.
- The total approved budget for the NGLV project is Rs. 8,240 crore. It will have three development flights with a target of 8 years for the completion of the development phase.
- The Mission also aims to achieve the goal of Reusable Launch Vehicle (RLV).



The PSLV-C62 rocket lifts off from Sriharikota on Monday. PTI

- On January 12, 2026, the ISRO's PSLV-C62 mission lifted off from Sriharikota, carrying the EOS-N1 satellite along with 15 co-passenger satellites. Within minutes, ISRO said the mission had encountered an anomaly during the end of the PS-3 stage.
- As per ISRO Chairman, the performance was as expected up to near the end of the rocket's third stage, called PS3. Then there was increased disturbance in the vehicle roll rates followed by a deviation in the flight path. In other words, towards the end of the third stage, the rocket was spinning uncontrollably enough for it to not be able to continue on its planned path.
- The rocket and the satellites were expected to fall back and burn up over the southern Indian Ocean.

The anomaly resembled the events preceding the failure of the PSLV C-61 mission on May 18, 2025.

In 2018, ISRO transitioned to using indigenous rubidium atomic clocks, developed by the ISRO-Space Applications Centre. The constellation has been degrading faster than it can be replenished, due to ISRO's poor launch rate. Now ISRO plans to launch three more second generation satellites in 2026.



- ISRO's NavIC constellation, for which it has launched 11 satellites since 2013, is in operational distress. Only three satellites remain capable of providing Position, Navigation, And Timing (PNT) services, leaving the constellation unable to fulfill its purpose of replacing the US's GPS system over the Indian subcontinent.
- A PNT constellation requires at least four PNT-capable satellites, and India had only four until ISRO said an atomic clock onboard the IRNSS-1F satellite failed on March 13.
- The constellation's first-generation satellites used rubidium atomic clocks made by Swiss company SpectraTime, and which have been dogged by failure.
- IRNSS-1F, launched in March 2016, completed its 10-year design life just three days before its clock failed. Eight other satellites have either been decommissioned, have failed to reach orbit or have bad clocks.

Part of NavIC's genesis was the US refusal to share GPS data over Kargil during the 1999 war. It continues to function largely as a defence program managed by ISRO.

What do you know about the BlueBird Block-2 satellite launched by ISRO recently?



- The Indian Space Research Organisation (ISRO) placed the BlueBird Block-2 satellite in a low earth orbit on December 24, 2025. The Launch Vehicle Mark-3 (LVM3) lifted off from the second launch pad of the Satish Dhawan Space Centre in Sriharikota.
- Two milestones have been accomplished by ISRO. It is the largest commercial communications satellite in low earth orbit. Weighing 6,100 kg, it became the heaviest payload to be launched by LVM3 from Indian soil.
- This is the first dedicated commercial launch for a customer from the USA, launched by the LVM3 Bahubali rocket-M6 launch vehicle.
- The satellite was developed by AST SpaceMobile in the US and is part of the next generation of BlueBird Block-2 communication satellites, designed to provide space-based cellular broadband connectivity directly to standard mobile smartphones.

The performance of the orbit is less than 2 kilometers and is one of the best performances of any launch vehicle in the global arena.

What do you know about GSAT-7R, India's heaviest communication satellite?



- ISRO successfully launched the Indian Navy's advanced communication satellite GSAT-7R (CMS-03) from the Satish Dhawan Space Centre in Sriharikota.
- The indigenously designed and developed satellite, weighing approximately 4,400 kg, is India's heaviest communication satellite launched from Indian soil so far and marks a major milestone in strengthening the Navy's space-based communications and maritime domain awareness.
- This satellite was launched aboard India's most powerful launch vehicle, the LVM3, on its M5 mission. The satellite had been successfully inserted into a Geosynchronous Transfer Orbit (GTO).

It will provide seamless connectivity between the navies, ships, submarines, aircraft, and maritime operations centers. It would provide robust and secure telecommunication coverage across the Indian Ocean region

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